Molarity and Dilution

Where $M_1V_1 = M_2V_2$ comes from, and what stops n ÷ V from being the whole story.

Name ______________________ Date ______________________ Period ______________________

Demonstration: andresdbp.github.io/resources/demos/molarity-dilution/

1 Before you open anything

Answer from what you already know. Being right is not the point — having something on paper to argue with later is.

PREDICTION

A beaker holds 0.100 L of $KMnO_4$ solution at 0.400 mol L^{-1}. You pour in 0.300 L of pure water. Predict the new concentration, in mol L^{-1}, and the number of moles of $KMnO_4$ in the beaker afterwards.

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Write one sentence saying how you got those two numbers.

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2 Building a solution from solid

Open the demonstration and stay on the **Make a solution** tab. Two sliders set what is in the beaker; the tiles underneath report the result five ways. Clicking a slider and then using the arrow keys is the reliable way to land on an exact value.

1. Press **Reset**. The sliders should read **0.100 mol** and **0.500 L**.
2. Set **Solute added, n** and **Volume of solution, V** to the values in the first two columns, one row at a time.
3. Record what **Molarity in solution**, **Dissolved**, **Undissolved** and **Percent of the limit** read for each row.

n (mol)	V (L)	Molarity in solution (mol L^{-1})	Dissolved (mol)	Undissolved (mol)	Percent of the limit
0.040	0.500				
0.100	0.500				
0.100	0.250				
0.200	0.500				
0.200	0.250				

a. In four of the five rows, Molarity in solution is exactly n ÷ V. Name the row where it is not, and write what the **Undissolved** tile reads there.

b. For that row, n ÷ V comes to 0.800 mol L^{-1}. The beaker does not. Account for every mole you added — how much is in solution, how much is not, and where the second amount is.

c. With n = 0.200 mol, **Mass weighed out** reads 31.61 g. Show the arithmetic that produces it, and state where the demonstration tells you the molar mass it used.

> The ceiling of 0.40 mol L^{-1} is not a property of arithmetic. It comes from 6.4 g of $KMnO_4$ per 100 mL of water **at 20 °C**. Warm the water and more dissolves; this model holds the temperature fixed, so the ceiling never moves.

3 What adding water does, and does not do

Switch to the **Dilute it** tab. The first two sliders decide how much solute goes into the beaker; the third adds pure water to it.

1. Set **Stock concentration, M_1** to **0.400 mol L^{-1}** and **Stock volume taken, V_1** to **0.100 L**.
2. Press **No water yet** and fill in the first row.

3. Move **Water added** to each value in the table and fill in the rest. For the last row you can press **Fill to 1.000 L** instead of dragging.

4. The last column is the second of the two equation lines, the one beginning M_2V_2. Copy the number it ends with.

Water added (L)	**Moles of solute in the beaker (mol)**	**Total volume now, V_2 (L)**	**Concentration now, M_2 (mol L^{-1})**	**Dilution factor**	**M_2V_2 line**
0					
0.100					
0.300					
0.400					
0.900					

a. Two columns hold the same value in every row. Which two, and why can neither of them change when water goes in?

b. Between row 1 and row 3, V_2 is multiplied by 4. What is M_2 multiplied by? State the relationship between the two factors in your own words, as a sentence about moles rather than as a formula.

c. Now turn that sentence into symbols. Write the equation it gives you, and say which one of the four quantities in it is the one you actually kept constant.

d. Go back to your prediction in part 1. Which of your two numbers held up? If either did not, name the assumption behind it.

4 Running it backwards

A question rarely hands you the answer volume. It hands you the solution you need and asks what to measure out. Work each one on paper *first*, then set the three sliders and see whether the readouts agree.

1. Target: 0.250 L of 0.0800 mol L^{-1}, made from 0.400 mol L^{-1} stock. Calculate V_1, then the volume of water needed.
2. Target: 0.500 L of 0.100 mol L^{-1}, made from 0.250 mol L^{-1} stock. Same two calculations.
3. Set the sliders to your answers and check **Concentration now, M_2** and **Total volume now, V_2**.

Target	V_1 you calculated (L)	Water to add (L)	M_2 the demonstration shows	V_2 it shows
0.250 L of 0.0800 M from 0.400 M				
0.500 L of 0.100 M from 0.250 M				

a. In both rows, the volume of water you add is not V_2 and not $V_2 \div V_1$. Say exactly what it is, and why the difference matters when you are standing at a bench with a graduated cylinder.

b. Next target: 0.500 L of 0.600 mol L^{-1} $KMnO_4$, made from solid. n ÷ V says you need 0.300 mol. Go to **Make a solution**, set V to **0.500 L** and push **Solute added, n** as high as the slider goes. Record **Molarity in solution** and **Percent of the limit**.

Give two separate reasons this target cannot be met in this model — one about the slider, one about the solution itself.

c. Could you reach 0.600 mol L^{-1} by starting from the most concentrated stock available in **Dilute it** and adding water? Answer in one sentence, and say what that tells you about the direction dilution can work in.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working, including units. Then reopen it, set the sliders to match, and check yourself. Take the molar mass of $KMnO_4$ as 158.03 g mol^{-1} and the solubility limit as 0.40 mol L^{-1} at 20 °C.

1. 4.74 g of $KMnO_4$ is dissolved and the solution is made up to 0.250 L. Calculate the molarity.

2. You need 0.500 L of 0.0400 mol L^{-1} $KMnO_4$ and you have 0.400 mol L^{-1} stock. Calculate the volume of stock to measure out and the volume of water to add.

3. 0.150 mol of $KMnO_4$ is added to water and made up to 0.200 L at 20 °C. Someone reports the concentration as 0.750 mol L^{-1}. Explain why that figure is unobtainable, then give the actual molarity, the moles that dissolve, and the mass of solid left on the bottom.

4. Without using $M_1V_1 = M_2V_2$, explain why adding 0.300 L of water to 0.100 L of *any* unsaturated stock divides its concentration by exactly 4, whatever that concentration was.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- In **Dilute it**, the plot draws a shaded rectangle and labels its area. Find two very different water settings whose rectangles look nothing alike, and check what the two areas come to. Then say what the dashed rectangle is doing there.
- The legend says each dot is 0.004 mol of dissolved $KMnO_4$. Count the dots at a few settings in both tabs and see whether the count follows the moles or the molarity. Watch what happens to the dots, and to the crystals, when you cross into saturation.
- In **Make a solution**, the plot has a bend where the straight line meets the solubility limit. Move the V slider across its whole range and track where that bend sits. Work out the rule for its position before you confirm it.
- Read the **Honest limits** panel. It admits that volumes are treated as additive. Explain how a volumetric flask filled to a mark sidesteps that assumption entirely.